AL-AZHAR UNIVERSITY

# Human Growth and Development

## Complete lifespan course for nursing students

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Public Health | Mother and Child Health

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# Course purpose

## Students will learn to

- Explain growth and development from conception through older adulthood
- Recognize expected patterns, variation, risk factors, and warning signs
- Apply developmental knowledge during assessment, communication, teaching, and care planning

## The nursing focus

- Observe function in context, not age alone
- Partner with the patient and family while respecting autonomy
- Use surveillance, validated screening, referral, and follow-up appropriately

MODULE 1

# Foundations of human development

Core concepts, patterns, determinants, and nursing assessment

# Growth and development

## Growth

- Quantitative change in body size or tissue mass
- Measured with length or height, weight, head circumference, body composition, and organ changes
- Best interpreted as a trend across time

## Development

- Change in function, competence, and adaptation
- Includes motor, language, cognitive, social, emotional, moral, and identity development
- Reflects maturation, experience, relationships, and environment

# Pediatric Growth Measurement

Accurate anthropometry, growth charts, interpretation, and nursing documentation

Nursing student module

MEASURE

**Weight**
**Length or height**
**Head circumference**

**Plot the result**
**Review the trend**
**Act on concerns**

# Infant Weight by Age (kg)

| Age | Boys | Girls |
|---|---|---|
| Birth | 3.3 | 3.2 |
| 6 months | 7.9 | 7.3 |
| 12 months | 9.6 | 8.9 |

WHO 50th-percentile values: reference medians, not required targets.

Source: WHO Child Growth Standards percentile tables, reproduced by KinderCure (2026).

# Infant Length by Age (cm)

| Age | Boys | Girls |
|---|---|---|
| Birth | 49.9 | 49.1 |
| 6 months | 67.6 | 65.7 |
| 12 months | 75.7 | 74.0 |

Use recumbent length (lying down) for infants, rather than standing height.

Source: WHO Child Growth Standards percentile tables, reproduced by KinderCure (2026).

# Head Circumference by Age (cm)

| Age | Boys | Girls |
|---|---|---|
| Birth | 34.5 | 33.9 |
| 6 months | 43.3 | 42.2 |
| 12 months | 46.1 | 44.9 |

Head circumference is a direct measurement; assess its trajectory on a growth chart.

Source: WHO Child Growth Standards percentile tables, reproduced by KinderCure (2026).

# Growth Calculations & Interpretation

Weight gain = later weight − earlier weight.
Example (boy): 7.9 − 3.3 = 4.6 kg from birth to 6 months.

Length gain = later length − earlier length.
Example (girl): 74.0 − 49.1 = 24.9 cm from birth to 12 months.

Head circumference gain = later measurement − earlier measurement.
Example (boy): 46.1 − 34.5 = 11.6 cm in the first year.

These examples compare reference medians, not an individual growth forecast.
Plot actual measurements by age and sex; the trend matters more than one number.

# Core growth measures across childhood

| Measure | Usual age or situation | What it contributes |
|---|---|---|
| **Weight** | Birth through adolescence | Change in body mass and response to illness or nutrition |
| **Recumbent length** | Birth to 24 months, or when the child cannot stand | Linear growth while lying supine |
| **Standing height** | From about 2 years when reliable standing is possible | Linear growth in an upright position |
| **Head circumference** | Especially birth to 24 months; follow local schedule | Growth of the skull and brain |
| **Weight for length or BMI for age** | Select according to age and chart system | Body proportionality and weight status screening |
| **MUAC** | Nutrition screening, often 6 to 59 months | Rapid screening for acute malnutrition in defined programs |

# Measurement quality starts before contact

## A small technique error can create a false change in the growth trajectory.

| | | |
|---|---|---|
| 1 | **Prepare** | Use the correct equipment. Place it on a firm, level surface. |
| 2 | **Check** | Inspect, calibrate, and zero the device according to policy. |
| 3 | **Explain** | Tell the caregiver and child what will happen. Protect privacy and warmth. |
| 4 | **Position** | Use the standard body position. Ask for assistance when required. |
| 5 | **Record** | Read promptly in metric units. Document date, age, and measurement method. |

# Infant weight and recumbent length



INFANT TECHNIQUE

## Weight

- Remove heavy clothing and the diaper when policy permits
- Place the infant in the centre of a tared infant scale
- Keep one hand close for safety without touching during the reading

## Recumbent length

- Use an infantometer and two trained measurers
- Hold the head against the fixed board
- Straighten both legs and bring the footboard firmly to the heels

# Standing weight and height

## Weight

- Use light clothing and no shoes
- Centre the child on the scale platform
- Wait for a stable reading
- Record immediately in kilograms

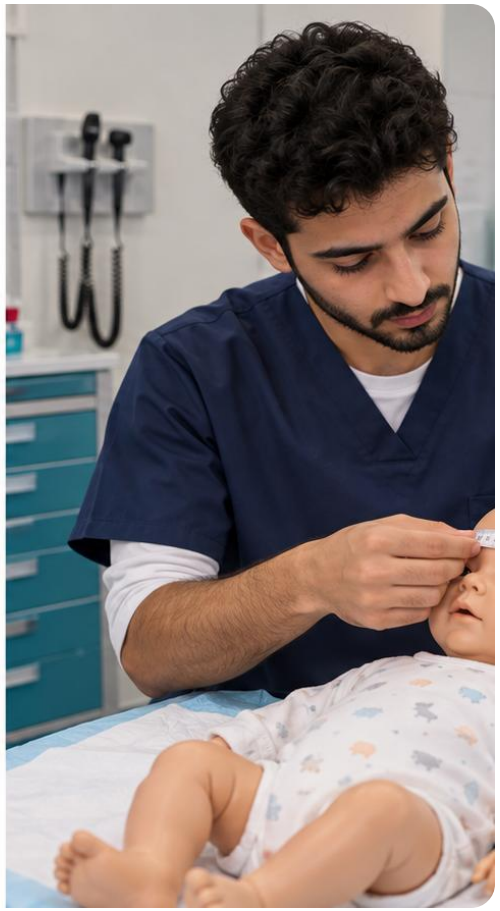
**Stay close enough to prevent a fall.**

## Height posture

| | |
|---|---|
| FEET | Bare and flat; heels together |
| BODY | Legs straight; shoulders relaxed |
| HEAD | **Frankfort plane horizontal** |
| HEADPIECE | Lower firmly onto the crown |
| READING | Read at eye level after full inhalation |

# Head circumference and MUAC





## Head circumference

- Use a flexible, nonstretch tape
- Place it above the eyebrows and ears
- Pass it around the largest occipital prominence
- Repeat and document the largest accurate measurement

## Mid upper arm circumference

Use MUAC only with the age range, tape, thresholds, and referral pathway specified by the local nutrition protocol.

# Selecting and plotting a growth chart

**Use the chart system adopted by the health service and select the correct age, sex, and indicator.**

### BIRTH TO 2 YEARS

Weight for age
Length for age
Weight for length
Head circumference

### 2 TO 5 YEARS

Weight and height
BMI for age or weight for height
Follow national policy

### 5 TO 19 YEARS

Height for age
BMI for age
WHO 5–19 reference where adopted

**Plot exact age and measurement, then compare with previous points**

# Interpreting the growth pattern

**Start with the trend, then decide whether the measurement and clinical story agree.**

## Patterns that need attention

- A confirmed value below −2 z scores or above +2 z scores, interpreted for the specific indicator
- A downward or upward change across major centile or z score lines
- Disproportion among weight, length or height, BMI, and head circumference
- Growth that conflicts with feeding history, illness, development, or physical assessment

**NURSING CHECK**

1. Confirm age and chart
2. Repeat an unexpected measurement
3. Review previous measurements
4. Assess history and clinical findings
5. **Document and escalate concerns**

**Growth charts support clinical assessment. They do not establish a diagnosis by themselves.**

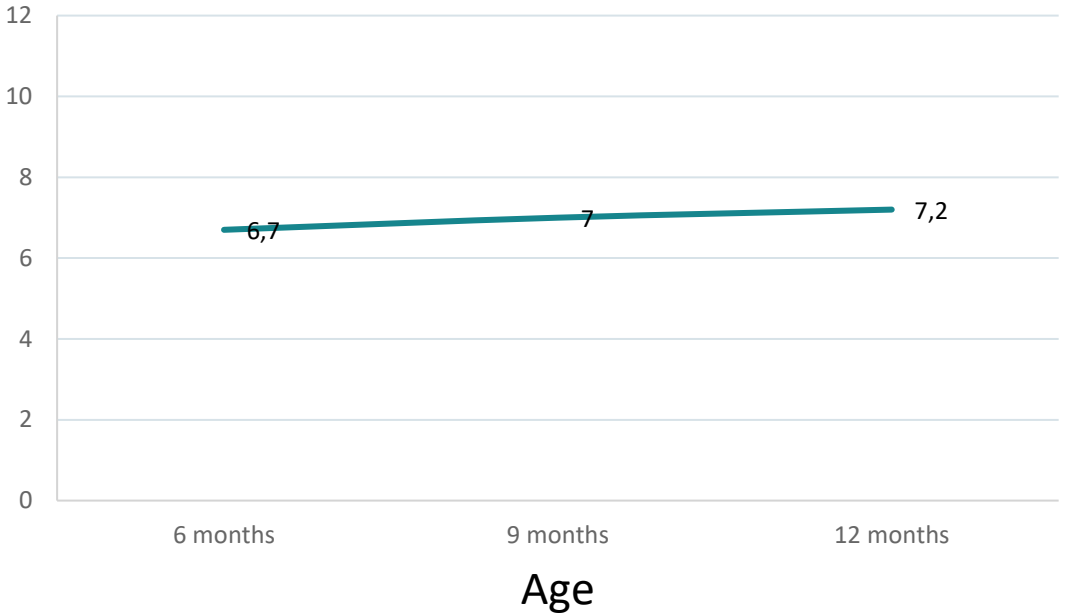
# Case activity: a flattening weight trend

## Teaching case: Samir, 12 months

Weight: 6 months 6.7 kg | 9 months 7.0 kg | 12 months 7.2 kg
Length at 12 months: 72.5 cm | Head circumference: 44.5 cm
Caregiver reports recent diarrhoea. Clothing used during weighing was not documented.

### Discuss in groups

- What measurement should the nurse verify first?
- Which chart and indicators are needed?
- What history and physical assessment data matter?
- What should the nurse document and report?




**The values are fictional and support skills practice. Do not diagnose from this graph alone.**

**Measure accurately. Plot correctly. Interpret the pattern. Act safely.**

# A lifespan and life-course view

1

## Biology

Genes, epigenetics, maturation, disease, and ageing shape capacity.

2

## Experience

Learning, attachment, education, work, and stress alter trajectories.

3

## Context

Family, culture, income, policy, and environment affect opportunity.

4

## Time

Earlier exposures can influence later health, but adaptation remains possible.

# Principles of growth and development

- Development continues across the lifespan and builds on earlier experience
- Order is often predictable, while timing and pace vary between people
- Development proceeds from broad responses toward more specific control
- Motor control generally follows cephalocaudal and proximodistal patterns
- Domains influence one another, so one concern may affect several functions
- Periods of rapid change can increase both opportunity and vulnerability

**A milestone is a guide to observation, not a stand-alone diagnosis.**

# Determinants of developmental health

## Protective influences

- Responsive relationships and emotional safety
- Adequate nutrition, sleep, movement, and preventive care
- Education, play, social connection, and accessible services
- Stable housing and protection from violence

## Risks and stressors

- Genetic or congenital conditions and prematurity
- Maternal illness, infection, malnutrition, toxins, or substance exposure
- Poverty, displacement, discrimination, or caregiver stress
- Chronic illness, pain, trauma, and social isolation

# The nursing assessment lens

1

## Patient

Current function, strengths, symptoms, sensory needs, and developmental history.

2

## Family

Caregiving, expectations, routines, health literacy, and stated concerns.

3

## Environment

Home, school, work, community, safety, access, and cultural context.

4

## Trajectory

Change over time, loss of skills, response to support, and follow-up.

# Developmental theories in nursing practice

| Framework | Central idea | Clinical use |
|---|---|---|
| Erikson | Psychosocial tasks continue across life | Anticipate needs related to trust, identity, intimacy, purpose, and meaning |
| Piaget | Thinking changes in structure and complexity | Match explanations to how the child understands causality and time |
| Attachment | Reliable care supports regulation and exploration | Observe caregiver-child interaction without blaming families |
| Bronfenbrenner | Development occurs within interacting systems | Assess family, school, community, services, and policy barriers |
| Life-course perspective | Timing and accumulated exposures shape health | Link present function to earlier risks and current resources |

MODULE 2

# Prenatal development and the newborn

Formation, adaptation, assessment, family teaching, and early escalation



# Prenatal development

1

## Germinal

Conception to about 2 weeks. Cell division, transport, and implantation.

2

## Embryonic

Weeks 3 to 8. Major structures form; harmful exposures can have large effects.

3

## Fetal

Week 9 to birth. Growth, differentiation, and functional maturation continue.

# Prenatal risks and protective care

## Assessment priorities

- Maternal health, nutrition, medications, infection, and substance exposure
- Gestational age, fetal growth, placental function, and multiple pregnancy
- Access to antenatal care, safety, mental health, and social support

## Nursing actions

- Use non-judgemental screening and clear health education
- Support nutrition, immunization, infection prevention, and medication review
- Recognize urgent symptoms and coordinate timely referral

# Newborn transition

- Respiratory and circulatory changes begin immediately after birth
- Thermoregulation, glucose control, feeding, and infection protection remain immature
- Behavioural states influence feeding, interaction, and examination findings
- Gestational age and birth history guide interpretation of size and function
- Skin-to-skin care supports warmth, regulation, bonding, and breastfeeding when clinically appropriate

**Observe the newborn before touching: colour, tone, breathing, posture, and responsiveness.**

# Focused newborn assessment

| Domain | Expected focus | Escalate when |
|---|---|---|
| Breathing and colour | Regular effort may include brief periodicity | Persistent central cyanosis, grunting, marked retractions, apnoea |
| Temperature and glucose risk | Warmth and feeding support | Temperature instability, jitteriness, lethargy, poor feeding |
| Neurologic state | Tone, symmetry, reflexes, responsiveness | Seizure, marked hypotonia, asymmetric movement |
| Feeding and hydration | Latch or suck, swallowing, urine and stool pattern | Weak suck, repeated vomiting, dehydration, bilious emesis |
| Jaundice and infection | Timing, extent, behaviour, risk factors | Jaundice in first 24 hours, fever or hypothermia, reduced responsiveness |

# Newborn teaching and safety

## Parents need to know

- Feeding cues and where to get feeding support
- Cord, skin, and temperature care
- Safe sleep and smoke-free surroundings
- Normal crying and safe soothing

## Seek urgent care for

- Difficulty breathing, blue colour, seizure, or reduced responsiveness
- Fever or abnormally low temperature
- Poor feeding, fewer wet nappies, repeated vomiting, or worsening jaundice
- Any caregiver concern that the baby is becoming less well

# Case: a sleepy 3-day-old newborn

PATIENT SNAPSHOT

A term newborn is difficult to wake for feeds. The mother reports fewer wet nappies today and increasing yellow colour. The baby feels cool. No temperature has been recorded.

## Clinical reasoning questions

- Which findings require immediate assessment?
- What focused history and measurements come first?
- How will you communicate urgency without blaming the parents?
- What documentation and referral steps are required?

MODULE 3

# Infancy and early childhood development

Growth measurement, nurturing care, developmental surveillance, milestones, play, safety, and early action



# Accurate growth measurement

1

## Prepare

Use calibrated equipment, minimal clothing, and the correct technique for age.

2

## Measure

Record weight, recumbent length or standing height, and head circumference when indicated.

3

## Plot

Use the appropriate sex- and age-specific standard or reference.

4

## Interpret

Assess the pattern, measurement quality, clinical context, and previous values.

# Interpreting a growth chart

## Look for

- A consistent trajectory rather than a single isolated percentile
- Concordance among weight, length or height, BMI or weight-for-length, and head growth
- Changes after illness, feeding difficulty, or altered family circumstances
- Expected variation related to gestational age and parental stature

## Check before concluding

- Age and gestational-age correction are accurate
- Equipment and technique were correct
- The correct chart and sex were selected
- Previous values are reliable and clinical assessment supports the pattern

# Infancy: selected milestones

| Age | Social and language | Motor and cognitive |
|---|---|---|
| 2 months | Calms when spoken to; looks at faces; sounds other than crying | Holds head up when prone; watches movement |
| 4 months | Smiles to gain attention; chuckles; turns toward voice | Holds head steady; brings hands to mouth; grasps a toy |
| 6 months | Knows familiar people; laughs; takes turns making sounds | Rolls from tummy to back; leans on hands when sitting; explores objects by mouth |
| 9 months | Responds to name; shows several facial expressions; babbles | Sits without support; transfers objects; looks for a dropped object |
| 12 months | Waves; calls a parent by a special name; understands “no” | Pulls to stand; cruises; pincer grasp; places an object in a container |

# Toddler: selected milestones

| Age | Communication and social | Movement, play, and thinking |
|---|---|---|
| 15 months | Shows affection; points to request; tries one or two words beyond names | Takes steps; finger-feeds; stacks two objects |
| 18 months | Points to show interest; follows a one-step direction; tries three or more words | Walks alone; scribbles; imitates household tasks |
| 2 years | Combines at least two words; notices when others are hurt | Runs; kicks a ball; uses switches or buttons; plays with more than one toy |
| 30 months | Says about 50 words; names things in a book; plays beside and sometimes with peers | Jumps with both feet; uses simple pretend play; follows two-step routines |
| 3 years | Back-and-forth conversation; asks who, what, where, or why; joins other children | Draws a circle when shown; strings items; uses a fork; puts on some clothes |

# Preschool: selected milestones

| Age | Communication and social | Movement, self-care, and thinking |
|---|---|---|
| 4 years | Pretend play; asks to play; comforts others; speaks in sentences of four or more words | Catches a large ball; serves food with supervision; draws a person with several body parts |
| 5 years | Follows rules in games; tells a simple story with events; keeps conversation going | Hops on one foot; buttons some buttons; counts to 10; writes some letters in own name |

# Health promotion in early childhood

## Developmental support

- Responsive caregiving and shared reading
- Daily active play and opportunities to explore safely
- Predictable routines with age-appropriate limits
- Language-rich interaction in the family's preferred language

## Safety and health

- Nutrition, oral health, immunization, sleep, and smoke-free air
- Age-appropriate restraints and protection from falls, burns, choking, drowning, and poisoning
- Screen caregiver well-being and barriers to care
- Teach families when and how to seek help

# Developmental warning signs

## Act promptly when

- A child loses a skill at any age
- The parent, caregiver, or nurse has a persistent concern
- Movement, communication, social interaction, hearing, vision, or feeding differs markedly from expectations
- Growth changes rapidly or crosses major channels

## The next steps

- Confirm history, observation, and measurement quality
- Use a validated screening tool when appropriate
- Assess hearing, vision, health, and social context
- Refer early and arrange follow-up; do not wait for failure to become severe

# Surveillance, screening, and diagnosis

| Approach | What it does | Nursing responsibility |
|---|---|---|
| Developmental surveillance | Ongoing observation, history, caregiver concerns, strengths, and trajectory | Perform at every relevant contact and document concerns |
| Standardized screening | Uses a validated tool at recommended ages or when concern arises | Administer correctly, score, explain limits, and follow the pathway |
| Diagnostic assessment | Determines whether a disorder or condition is present | Refer to qualified professionals and support the family during evaluation |

# Case: an 18-month-old with few words

PATIENT SNAPSHOT

An 18-month-old walks independently and enjoys spinning the wheels of a toy car. The parent reports no clear words, limited pointing, and inconsistent response to name. Growth is steady.

## Clinical reasoning questions

- Which strengths and concerns will you document?
- What hearing, developmental, and social-communication questions are needed?
- Which screening and referral pathway fits this setting?
- How will you explain early referral in supportive language?

# Early childhood development: nursing scope

- Early childhood includes the period from birth through the transition to school, with rapid change in brain, body, language, relationships, and self-regulation
- Nurses observe development during routine care, illness visits, immunization contacts, home visits, and community work
- Assessment includes caregiver concerns, child strengths, growth, milestones, behaviour, hearing, vision, nutrition, safety, and social context
- Support begins before a diagnosis through responsive caregiving, play, communication, nutrition, caregiver support, and early referral
- Repeated observation and closed-loop follow-up matter more than a single checklist

**Every contact with a young child can strengthen development or identify a need for early support.**

# Nurturing care in early childhood

1

## Health

Preventive care, timely treatment, immunization, and management of disability or chronic illness.

2

## Nutrition

Age-appropriate feeding, growth monitoring, micronutrients, and support for feeding difficulties.

3

## Responsive care

Notice the child's cues, respond consistently, and support secure relationships.

4

## Safety

Protect from injury, violence, neglect, toxic stress, and unsafe environments.

5

## Early learning

Conversation, shared reading, play, exploration, and everyday routines that build skills.

# Responsive caregiving and early learning

## What caregivers can do

- Follow the child's attention and respond to sounds, gestures, words, and emotions
- Talk during feeding, bathing, dressing, and travel using the family's preferred language
- Share books, songs, imitation games, and simple pretend play
- Offer safe choices and praise effort rather than demanding perfect performance

## What nurses observe

- Shared attention, eye contact that fits the child, turn-taking, comfort, and enjoyment
- How the child signals needs and how the caregiver responds
- Opportunities for movement, play, language, and peer interaction
- Caregiver stress, depression, safety concerns, and barriers to responsive care

# Play and communication opportunities

| Developmental period | Helpful caregiver activity | Nursing observation |
|---|---|---|
| Infant | Face-to-face talk, songs, floor play, reaching and safe exploration | State regulation, social response, movement symmetry, hearing and vision |
| Toddler | Naming, pointing, simple choices, imitation, blocks, and active play | Joint attention, words and gestures, walking, problem solving, caregiver response |
| Preschool | Pretend play, stories, drawing, turn-taking, and simple household tasks | Conversation, imagination, peer interest, fine motor skill, self-care |
| School transition | Routines, shared reading, cooperative games, and growing responsibility | Readiness to participate, follow directions, communicate needs, and manage emotion |

# Practical ECD observation during a nursing visit

1

## Watch

Observe arrival, caregiver-child interaction, spontaneous movement, play, and communication.

2

## Ask

Invite caregiver concerns and ask what the child can do at home in familiar routines.

3

## Engage

Offer a simple age-appropriate toy, book, or task without coaching the answer.

4

## Document

Record strengths, concerns, tools used, result, action, referral owner, and follow-up date.

# ECD screening, documentation, and referral

## Use tools correctly

- Choose a validated tool appropriate to age, language, setting, and purpose
- Explain that screening estimates risk and does not diagnose a disorder
- Score according to the manual and never change cut-offs informally
- Combine the result with surveillance, clinical assessment, and caregiver concern

## Close the loop

- Discuss results in clear, supportive language and identify immediate safety or sensory needs
- Refer through the local pathway for hearing, vision, medical, developmental, or social evaluation
- Give families practical activities they can begin now
- Track whether the referral occurred and whether function changed

# Case: a 3-year-old with limited pretend play

PATIENT SNAPSHOT

A 3-year-old uses short phrases but rarely joins other children or uses pretend play. The caregiver reports long work hours, housing stress, and little access to safe play space. Hearing has not been tested recently.

## Clinical reasoning questions

- Which strengths, developmental domains, and environmental factors will you assess?
- What hearing, vision, developmental, and family-support actions are appropriate?
- Which practical play and communication activities could start today?
- How will you document referral responsibility and follow-up?

FOCUS MODULE

# Developmental domains in depth

Gross motor, fine motor, language, cognition, social-emotional development, adaptive function, and play from birth to 5 years

# Six developmental domains nurses observe

| Domain | What develops | Examples in practice |
|---|---|---|
| Gross motor | Posture, balance, large-muscle control, mobility | Head control, sitting, walking, running, jumping |
| Fine motor and visual-motor | Hand use, grasp, release, bilateral coordination | Reaching, pincer grasp, drawing, buttons |
| Language and communication | Understanding, expression, speech, gesture, social use | Response to sound, pointing, words, conversation |
| Cognitive | Attention, memory, learning, concepts, problem solving | Object search, cause and effect, pretend play, counting |
| Social-emotional | Relationships, emotional regulation, identity, participation | Social smile, joint attention, peer play, empathy |
| Adaptive and self-care | Everyday independence and participation | Feeding, dressing, toileting, routines, safety awareness |

# Developmental domains influence one another

1

## Regulation

Sleep, comfort, sensory processing, and emotional safety affect attention and learning.

2

## Movement

Postural control and hand use expand exploration, play, feeding, and communication.

3

## Communication

Gesture and language help the child request, share attention, learn, and regulate emotion.

4

## Participation

Relationships and accessible environments determine how skills appear in daily life.

# Motor development: foundations

## Neurologic and body systems

- Maturation of the central and peripheral nervous systems
- Muscle strength, tone, endurance, joint mobility, and skeletal growth
- Vision, vestibular function, touch, and proprioception
- Health, nutrition, sleep, pain, and cardiopulmonary capacity

## Experience and environment

- Safe floor time and freedom to move
- Opportunities to reach, grasp, climb, and play
- Responsive encouragement without forcing a skill
- Accessible spaces and supports for children with disabilities

# Posture, tone, balance, and coordination

- Postural control stabilizes the head and trunk so the child can look, reach, breathe, feed, and move efficiently
- Muscle tone describes resistance to passive movement and should be interpreted with posture, strength, symmetry, and function
- Balance develops through visual, vestibular, and somatosensory input
- Coordination links timing, force, direction, and sequencing across body segments
- Persistent asymmetry, unusual stiffness, marked floppiness, or loss of function needs prompt evaluation

**Observe quality of movement as well as whether the child completes the task.**

# From reflexive to voluntary movement

## Early motor organization

- Newborn movement includes spontaneous activity and age-linked reflex responses
- Head and trunk control gradually improve through antigravity movement
- Hands open more often and reaching becomes purposeful
- Righting, protective, and balance reactions support safe mobility

## Clinical interpretation

- Check gestational age and medical history before comparing performance
- Observe both sides and several positions
- Consider vision, hearing, pain, fatigue, and opportunity to practice
- Refer when reflexes, tone, symmetry, or movement quality are concerning

# Gross motor development

- Gross motor skills use the trunk and large muscle groups for posture and movement
- Control generally progresses from head toward feet and from the body centre toward the limbs
- Skills build through overlapping sequences rather than one rigid timetable
- Children may use different safe movement strategies before independent walking
- Milestone monitoring describes function, while standardized screening estimates risk

**Key questions: What can the child do, how do they do it, and how does it affect participation?**

# Gross motor observation in nursing care

| Observe | Look for | Document |
|---|---|---|
| Supine and prone | Head position, antigravity movement, symmetry, tolerance | Position, assistance, duration, movement quality |
| Pull to sit and supported sitting | Head lag, trunk control, protective responses | Level of support and alignment |
| Transitions | Rolling, moving into sitting, floor mobility, rising | Independent, assisted, preferred side |
| Standing and gait | Weight bearing, base, balance, symmetry, falls | Surface, footwear, distance, support |
| Play | Running, stairs, ball skills, jumping, hopping | Opportunity, confidence, endurance, participation |

# Gross motor profile: birth to 2 months

## Expected progression

- Flexed newborn posture gradually becomes less dominant
- Turns head in both directions when lying
- Moves both arms and both legs
- By 2 months, holds the head up briefly during supervised tummy time

## Nursing focus

- Observe spontaneous movement, tone, symmetry, and state regulation
- Teach awake, supervised tummy time in short frequent periods
- Review safe sleep separately from awake play
- Escalate poor feeding, marked asymmetry, unusual tone, or reduced movement

# Gross motor profile: 4 months

## Expected progression

- Holds the head steady when supported upright
- Pushes up on elbows or forearms while prone
- Brings hands toward the midline and mouth
- Uses arms to swing toward nearby toys

## Nursing focus

- Observe head control without pulling the infant into sitting
- Check symmetry of head position and limb movement
- Ask about floor play and time spent in restrictive equipment
- Consider torticollis, plagiocephaly, vision, pain, and tone when movement differs

# Gross motor profile: 6 months

## Expected progression

- Rolls from tummy to back
- Pushes up with straight arms while prone
- Leans on hands for support in sitting
- Reaches for a desired toy while maintaining balance

## Nursing focus

- Provide safe floor space for rolling and reaching
- Observe trunk control and use of both sides
- Avoid comparing one isolated skill without the whole profile
- Investigate loss of movement, persistent head lag, marked stiffness, or marked floppiness

# Gross motor profile: 9 months

## Expected progression

- Gets into a sitting position
- Sits without support
- Uses the floor to reach and explore
- May begin pulling toward standing or other supported mobility

## Nursing focus

- Observe transitions, not only the final sitting posture
- Ask how the infant moves across the floor at home
- Check hand transfer, vision, hearing, and social response with motor skills
- Inability to sit independently by 9 months warrants timely evaluation

# Gross motor profile: 12 months

## Expected progression

- Pulls to stand
- Walks while holding furniture
- Changes position to reach toys and people
- May stand or take steps independently, with wide normal variation

## Nursing focus

- Ask about safe cruising surfaces and fall hazards
- Observe weight bearing on both legs and foot position
- Do not use infant walkers
- Explore asymmetry, persistent toe standing, pain, or limited participation

# Gross motor: 15 to 18 months

## Expected progression

- At 15 months, takes a few steps independently
- By 18 months, walks without holding on
- Climbs onto and off a low chair or couch
- Begins faster walking and early running attempts

## Nursing focus

- Observe gait, balance, transitions, and falls
- Ask whether mobility is changing or has stopped progressing
- Encourage active play in safe open spaces
- No independent walking by 18 months requires developmental and medical review

# Gross motor profile: 2 years

## Expected progression

- Runs
- Kicks a ball
- Walks up a few stairs with or without help
- Changes direction and explores climbing with improving control

## Nursing focus

- Observe safety awareness, coordination, and endurance
- Ask about frequent falls, pain, fatigue, and participation
- Promote active play and supervised practice on varied safe surfaces
- Assess vision, hearing, neurologic findings, and musculoskeletal alignment when concerned

# Gross motor profile: 30 months to 3 years

## Expected progression

- At 30 months, jumps with both feet off the ground
- Runs and stops with improving control
- Uses stairs with increasing independence
- Participates in active play with peers

## Nursing focus

- Observe the child during free movement before structured tasks
- Ask about playground access and opportunities for practice
- Explore fear, sensory needs, pain, weakness, and breathing limits
- Consider referral when movement restricts self-care or peer participation

# Gross motor profile: 4 to 5 years

## Expected progression

- At 4 years, catches a large ball most of the time
- Balances, climbs, and changes direction with greater control
- By 5 years, hops on one foot
- Joins rule-based active games and manages more complex sequences

## Nursing focus

- Ask about confidence and participation rather than athletic performance
- Observe coordination, balance, endurance, and safety judgement
- Screen vision and hearing when school or movement concerns arise
- Support inclusive activity for children with chronic conditions or disabilities

# WHO gross motor windows of achievement

- WHO describes population windows for sitting without support, standing with assistance, hands-and-knees crawling, walking with assistance, standing alone, and walking alone
- The windows show normal variation across healthy children rather than a single deadline
- Crawling is one possible mobility pattern and is not required before walking
- Interpret milestone timing with the child's trajectory, medical history, movement quality, and participation
- Use local clinical pathways when a skill falls outside expectations or when quality is concerning

**A window supports interpretation. It does not replace clinical judgement or standardized screening.**

# Gross motor warning signs

| Finding | Why it matters | Immediate nursing action |
|---|---|---|
| Loss of a previously acquired motor skill | May signal neurologic or systemic illness | Urgent medical evaluation based on acuity |
| Marked asymmetry or early persistent hand preference | May reflect unilateral weakness or injury | Document side, function, tone, and refer |
| Persistent head lag, marked stiffness, or marked floppiness | May indicate altered neuromotor control | Complete focused assessment and escalate |
| Not sitting independently by 9 months | Important developmental concern | Screen, assess associated domains, and refer |
| Not walking independently by 18 months | Needs developmental and medical evaluation | Check hearing, vision, tone, strength, and referral pathway |
| Pain, fatigue, breathing difficulty, or declining endurance | May represent acute or chronic disease | Assess vital signs, function, and medical cause |

# Nursing response to a motor concern

1

## Clarify

Confirm age, gestation, onset, progression, opportunities, and caregiver concerns.

2

## Observe

Watch spontaneous movement, posture, symmetry, transitions, gait, and participation.

3

## Examine

Assess tone, strength, reflexes, joints, vision, hearing, pain, and general health.

4

## Act

Use validated screening, refer through local pathways, teach safe activity, and track follow-up.

# Fine motor and visual-motor development

- Fine motor skills use the hands and fingers for reaching, grasping, releasing, manipulating, drawing, and self-care
- Visual-motor integration coordinates what the eyes see with how the hands act
- Bilateral coordination allows the two hands to stabilize, transfer, and work together
- Hand skills depend on posture, shoulder stability, sensation, vision, attention, and opportunity
- Nurses should observe function during play, feeding, dressing, and drawing

**Fine motor performance cannot be interpreted without the child's motor, sensory, cognitive, and environmental context.**

# Grasp and release progression

| Pattern | Description | Everyday example |
|---|---|---|
| Palmar contact | Object held against the palm with limited finger control | Holding a rattle placed in the hand |
| Purposeful reach | Arm and hand move toward a visible object | Reaching for a toy |
| Transfer | Object moves from one hand to the other | Exploring a block with both hands |
| Raking grasp | Fingers sweep a small object toward the palm | Collecting soft finger food |
| Pincer grasp | Thumb and index finger pick up a small object | Picking up a small safe food piece |
| Refined tool use | Grasp adapts to crayons, utensils, fasteners, and scissors | Drawing, eating, dressing |

# Visual-motor integration

## Skills to observe

- Fixes and tracks a face or toy
- Reaches accurately toward an object
- Adjusts hand shape before contact
- Uses one hand to stabilize while the other manipulates
- Copies lines, shapes, and later letter-like forms

## Factors that may affect performance

- Visual impairment or ocular-motor difficulty
- Postural instability or limited shoulder control
- Altered tone, sensation, pain, or joint movement
- Attention, comprehension, fatigue, or unfamiliar materials
- Limited opportunities for safe play and tool use

# Fine motor: 2 to 4 months

| Expected progression | Nursing focus |
| --- | --- |
| Opens hands briefly by 2 months | Observe both hands and arm movement |
| Looks at hands with interest | Offer a safe lightweight object at midline |
| Brings hands to mouth | Check vision and shoulder movement when reaching is limited |
| Holds a toy placed in the hand and swings toward toys | Persistent fisting, marked asymmetry, or reduced movement needs evaluation |

# Fine motor profile: 6 to 9 months

## Expected progression

- Reaches and grasps a desired toy
- Explores objects with hands and mouth
- Transfers objects between hands by 9 months
- Uses fingers to rake food toward the hand

## Nursing focus

- Use safe objects that are too large to choke on
- Observe reach accuracy, grasp, release, and hand transfer
- Check sitting balance because posture affects hand use
- Investigate lack of transfer, persistent asymmetry, or limited visual attention

# Fine motor profile: 12 to 15 months

## Expected progression

- Picks up small safe items between thumb and index finger
- Puts objects into a container
- Uses fingers to self-feed
- Stacks at least two small objects by 15 months

## Nursing focus

- Observe hand preference, reach, release, and bilateral use
- Ask about feeding textures and choking safety
- Use blocks, cups, and books for brief play observation
- Assess vision, tone, and sensory response when manipulation is difficult

# Fine motor profile: 18 to 24 months

## Expected progression

- Scribbles by 18 months
- Attempts spoon use and drinks from an open cup
- Uses one hand to hold while the other manipulates by 2 years
- Operates simple knobs, switches, and buttons on toys

## Nursing focus

- Observe utensil use without expecting neat performance
- Offer thick crayons and simple containers
- Ask how hand skills affect feeding and play
- Explore vision, weakness, tremor, pain, or loss of skill when concerned

# Fine motor profile: 30 months to 3 years

## Expected progression

- Turns book pages one at a time by 30 months
- Twists lids or doorknobs
- Draws a circle after demonstration by 3 years
- Strings large items and uses a fork

## Nursing focus

- Observe pencil grasp without forcing a mature pattern
- Use drawing, beads, puzzles, and self-care tasks
- Check comprehension before interpreting task failure
- Consider occupational therapy when hand skills limit participation

# Fine motor profile: 4 to 5 years

## Expected progression

- Uses a finger-thumb grasp for crayons or pencils by 4 years
- Draws a person with several body parts
- Unbuttons some buttons and serves food with supervision
- By 5 years, buttons some buttons and writes some letters in the name

## Nursing focus

- Ask about drawing, dressing, utensils, and classroom participation
- Observe visual attention, posture, and hand use
- Separate lack of opportunity from loss or inability
- Coordinate vision, developmental, and occupational therapy referral as indicated

# Fine motor warning signs

| Finding | Possible contributors | Nursing response |
|---|---|---|
| Persistent hand asymmetry in infancy | Weakness, pain, injury, altered tone | Compare sides, assess movement quality, refer |
| No transfer between hands near 9 months | Motor, visual, sensory, or opportunity factors | Observe posture and reach, screen, refer if persistent |
| Difficulty with pincer grasp or self-feeding after expected progression | Fine motor, oral-motor, sensory, or vision concern | Assess safety and daily function |
| Hand skills limit play, dressing, or school tasks | Coordination, strength, vision, cognition, environment | Document participation and seek multidisciplinary input |
| Loss of grasp, drawing, or self-care skill | Neurologic or systemic illness | Prompt medical evaluation |

# Promoting fine motor development safely

## Helpful opportunities

- Floor play with safe objects of different shapes and textures
- Blocks, containers, large beads, puzzles, paper, crayons, and play dough
- Self-feeding and dressing with patient adult support
- Drawing and construction activities linked to the child’s interests

## Safety and inclusion

- Avoid small choking hazards for children who mouth objects
- Adapt handles, seating, and task demands when needed
- Respect sensory preferences and allow extra response time
- Use play to observe strengths, not to pressure performance

# Language and communication: four components

| Component | Meaning | Examples |
|---|---|---|
| Receptive language | Understanding words, sentences, gestures, and meaning | Responds to name, follows directions, identifies objects |
| Expressive language | Communicating ideas through sounds, gestures, signs, or words | Babbling, pointing, naming, combining words |
| Speech | Motor production of voice sounds and intelligible words | Sound patterns, fluency, voice quality, clarity |
| Pragmatics | Using communication socially and appropriately | Turn-taking, joint attention, topic maintenance, repair |

# Foundations of language development

- Hearing gives access to speech and environmental sound, while vision supports faces, gestures, and shared attention
- Responsive turn-taking teaches the rhythm and purpose of communication
- Joint attention occurs when child and caregiver share focus on the same object or event
- Gestures such as reaching, showing, giving, pointing, and waving bridge early communication and words
- Play, imitation, memory, motor control, and relationships all support language

**A child can communicate meaningfully before speech develops. Observe sounds, gestures, gaze, signs, and actions.**

# Communication profile: birth to 6 months

## Expected progression

- Reacts to loud sound and calms to a familiar voice
- Makes sounds other than crying by 2 months
- Coos and makes sounds back during interaction by 4 months
- By 6 months, takes turns making sounds, squeals, and laughs

## Nursing focus

- Ask caregivers what sounds the infant notices
- Observe face-to-face turn-taking and response to voice
- Review newborn hearing results and later risk factors
- Refer promptly for absent sound response, loss of vocalization, or caregiver concern

# Communication profile: 6 to 12 months

## Expected progression

- Produces varied repeated syllable strings by 9 months
- Raises arms to request being picked up
- Looks when called by name
- By 12 months, waves, uses a special parent name, and understands a brief no

## Nursing focus

- Observe gestures, social response, and sound variety
- Ask about response to soft sounds and voices in daily routines
- Encourage imitation, shared books, songs, and naming
- Inconsistent response to name requires hearing and broader developmental review

# Communication profile: 12 to 18 months

## Expected progression

- Points to ask for help by 15 months
- Attempts one or two words beyond parent names
- Follows a direction paired with gesture and words
- By 18 months, attempts three or more words and follows a one-step direction without gesture

## Nursing focus

- Count communication functions, not only spoken vocabulary
- Ask about words and gestures in every language used at home
- Observe joint attention, imitation, and pretend play
- Combine hearing evaluation, developmental screening, and referral when concerns arise

# Communication profile: 2 years

## Expected progression

- Combines at least two words, such as more milk
- Points to named pictures and body parts
- Uses gestures beyond waving and pointing
- Communication increasingly supports requests, sharing, play, and emotion

## Nursing focus

- Ask whether the child understands familiar directions
- Observe spontaneous communication rather than prompted repetition alone
- Check hearing even when the child reacts to some sounds
- Refer for missing word combinations, loss of words, or limited social communication

# Communication profile: 30 months

## Expected progression

- Uses about 50 words
- Combines two or more words with an action word
- Names familiar pictures in a book
- Uses pronouns such as I, me, or we

## Nursing focus

- Listen for variety of words and communication purposes
- Ask how the child communicates needs with unfamiliar people
- Observe understanding of two-step instructions
- Refer when language limits safety, play, behaviour regulation, or participation

# Language: age 3 years

## Expected progression

- Uses at least two back-and-forth exchanges in conversation
- Asks who, what, where, or why questions
- Names an action shown in a picture
- States first name and is understood by others most of the time

## Nursing focus

- Use open questions and allow time to answer
- Observe comprehension separately from speech clarity
- Ask about communication across home, childcare, and play
- Assess hearing, oral-motor function, language, and social communication when concerned

# Communication profile: 4 years

## Expected progression

- Uses sentences with four or more words
- Recalls something that happened during the day
- Says parts of a song, story, or rhyme
- Answers simple questions about object use

## Nursing focus

- Invite a brief story about play or family routine
- Note sentence length, grammar, clarity, and conversational reciprocity
- Ask caregivers and teachers about comprehension and peer communication
- Refer when communication limits learning, relationships, or daily needs

# Communication profile: 5 years

## Expected progression

- Tells a story with at least two connected events
- Answers questions about a story
- Maintains more than three conversational exchanges
- Recognizes or produces simple rhymes

## Nursing focus

- Observe narrative organization and topic maintenance
- Ask about classroom listening and following group directions
- Check hearing and vision when learning concerns appear
- Coordinate speech-language and educational evaluation as appropriate

# Multilingual language development

- Exposure to more than one language does not cause a language disorder
- Evaluate skills across all languages the child understands and uses
- Vocabulary may be distributed across languages, so one-language counts can underestimate knowledge
- A true language difficulty generally affects learning and communication across the child’s languages
- Encourage families to speak, read, sing, and tell stories in the languages they use most comfortably
- Use trained interpreters and culturally appropriate validated tools

**Do not advise a family to stop using the home language as a response to delay.**

# Speech clarity and oral-motor function

## Observe speech

- Sound variety and consistency
- Voice quality, volume, resonance, and fluency
- How well familiar and unfamiliar listeners understand
- Whether frustration or avoidance affects participation

## Observe related function

- Oral structure and movement when clinically indicated
- Chewing, swallowing, drooling, and feeding safety
- Breathing, sleep, recurrent ear disease, and hearing
- Use of gesture or augmentative communication

# Language and communication warning signs

| Concern | Clinical meaning | Action |
|---|---|---|
| No response to sound or voice | Possible hearing, neurologic, or state-regulation concern | Hearing and medical evaluation |
| Limited gestures or joint attention | May affect social communication and language learning | Developmental surveillance, screening, and referral |
| No meaningful words by the expected toddler period | Language delay may be present | Assess hearing and all developmental domains |
| No two-word combinations by 2 years | Important expressive-language concern | Validated screen and timely referral |
| Loss of words, gestures, or social response | Developmental regression | Prompt medical and developmental evaluation |
| Speech or language restricts safety or participation | Functional impact requires support | Refer and begin communication supports now |

# Hearing, language, and referral

1

## History

Review newborn screening, infections, ototoxic exposure, family history, ear disease, and caregiver concern.

2

## Observe

Note response to voice, environmental sound, gesture, language, and visual cues.

3

## Test

Arrange age-appropriate hearing assessment when speech, language, or sound response is concerning.

4

## Support

Continue communication, early intervention, and family guidance while evaluation proceeds.

# Cognitive development

- Cognition includes attention, memory, learning, concepts, imitation, problem solving, symbolic thought, and executive function
- Infants learn through repeated sensory and motor experiences
- Toddlers test cause and effect and begin symbolic or pretend play
- Preschool children expand memory, classification, early number concepts, and planning
- Language, motor access, relationships, sleep, nutrition, hearing, vision, and stress affect cognitive performance

**Observe how the child approaches a task, not only whether the answer is correct.**

# Sensorimotor learning

## How infants learn

- Look, listen, touch, mouth, and move
- Repeat actions that produce an interesting result
- Imitate facial expressions, sounds, and actions
- Build expectations through predictable routines

## Nursing opportunities

- Offer safe objects for exploration
- Follow the infant’s attention and describe what is happening
- Repeat simple games with pauses for the infant to respond
- Reduce overload when the infant turns away, stiffens, or becomes distressed

# Object permanence and cause and effect

## Emerging concepts

- Looks for a dropped or partly hidden object
- Repeats an action to make something happen
- Uses containers, lids, switches, and simple tools
- Anticipates familiar routines and people

## Observation ideas

- Hide part of a toy under a cloth
- Offer a container with blocks to remove and replace
- Use a simple push-button toy
- Observe persistence, strategy changes, and request for help

# Toddler problem solving and pretend play

## Expected progression

- Uses familiar objects for their purpose
- Combines toys in simple sequences
- Pretends one object represents another
- Follows one-step and later two-step directions
- Tries a new strategy when the first attempt fails

## Nursing observation

- Offer a doll, cup, spoon, block, and simple puzzle
- Watch spontaneous play before demonstrating
- Ask caregivers about play at home and with peers
- Document imitation, flexibility, attention, and symbolic use

# Preschool executive function

- Working memory holds information long enough to follow a direction or continue a game
- Inhibitory control helps the child pause, wait, and resist an immediate impulse
- Cognitive flexibility supports shifting between rules, ideas, or perspectives
- These capacities develop gradually through play, routines, language, sleep, and co-regulation
- Stress, fatigue, hunger, unfamiliar settings, and developmental conditions can reduce performance

**Use simple directions and observe the support a child needs to succeed.**

# Cognitive warning signs

| Observation | Explore | Nursing action |
|---|---|---|
| Limited exploration or curiosity | Vision, hearing, motor access, mood, pain, opportunity | Adapt the environment and assess related domains |
| Little imitation or pretend play | Social communication, language, experience, cognition | Screen and refer when persistent |
| Cannot follow familiar directions | Understanding, hearing, attention, context | Clarify language and use validated screening |
| Difficulty solving simple age-linked tasks | Task familiarity, motor demands, stress, cognition | Observe strategies and functional impact |
| Loss of learned concepts or play skills | Regression or illness | Prompt medical evaluation |

# Social-emotional development

- Social-emotional development includes attachment, trust, emotional expression, regulation, empathy, identity, and relationships
- Young children regulate emotion with caregiver support before they can self-regulate consistently
- Temperament affects intensity, adaptability, and response to new experiences
- Culture shapes expectations for eye contact, independence, expression, sleep, feeding, and play
- Nursing assessment includes the child, caregiver-child relationship, and family stressors

**Describe behaviour in context and avoid labelling the child.**

# Attachment and co-regulation

## Caregiver contributions

- Notice cues and respond consistently
- Offer comfort and protection during distress
- Share attention, pleasure, and predictable routines
- Support exploration while remaining available

## Child contributions

- Signals needs through movement, expression, sound, and behaviour
- Seeks closeness during stress
- Uses the caregiver as a secure base for exploration
- Gradually learns strategies for calming and waiting

# Social-emotional profile: infancy

## Expected progression

- Looks at faces and smiles during interaction
- Shows pleasure with familiar people
- Expresses a wider range of emotions
- Responds to name and reacts when a caregiver leaves
- Enjoys interactive games such as peek-a-boo

## Nursing focus

- Observe mutual attention, comfort, and recovery from distress
- Ask about caregiver mood, stress, and support
- Consider hearing, vision, temperament, and medical state
- Respond early to limited social response or loss of interaction

# Social-emotional profile: toddler

## Expected progression

- Shows objects and points to share interest
- Moves away to explore while checking caregiver proximity
- Copies household actions and begins simple pretend play
- Notices when another person is upset
- Plays beside peers and gradually joins shared play

## Nursing focus

- Normalize brief frustration while assessing intensity and recovery
- Observe joint attention, imitation, and flexibility
- Ask about routines, sleep, family stress, and behaviour across settings
- Support clear limits, simple choices, and caregiver co-regulation

# Social-emotional profile: preschool

## Expected progression

- Joins other children in play
- Uses pretend roles and understands simple rules
- Comforts others and likes to help
- Adjusts behaviour across familiar settings
- Takes turns and completes simple household responsibilities

## Nursing focus

- Ask about peer relationships, childcare, and school readiness
- Observe conversation, imaginative play, and emotional recovery
- Explore anxiety, aggression, withdrawal, trauma, and family stress
- Promote predictable routines and positive attention

# Social-emotional warning signs

| Concern | Context to assess | Action |
|---|---|---|
| Limited social response or shared attention | Hearing, vision, interaction, communication, temperament | Developmental screening and referral |
| Persistent extreme distress or poor recovery | Pain, sleep, feeding, sensory needs, caregiver stress | Assess health and relationship supports |
| Little interest in people or play | Opportunity, depression, neurodevelopment, illness | Observe across settings and refer |
| Severe aggression or self-injury | Triggers, communication, trauma, safety, medical causes | Create a safety plan and escalate |
| Loss of social or play skills | Regression | Prompt medical and developmental evaluation |

# Adaptive development and self-care

- Adaptive skills allow the child to participate in feeding, dressing, hygiene, toileting, routines, safety, and community life
- Performance depends on motor, sensory, cognitive, communication, and social capacities
- Family expectations and culture influence when and how children practise independence
- Illness, disability, stress, poverty, and inaccessible environments may restrict opportunity
- Assess what the child does in familiar daily routines and what support makes participation possible

**Measure independence together with safety, dignity, and participation.**

# Feeding development

## Progressive skills

- Coordinates sucking, swallowing, and breathing in infancy
- Explores safe textures and self-feeds with fingers
- Drinks from an open cup with help and then increasing independence
- Uses spoon and fork with improving control
- Joins family meals and communicates hunger and fullness

## Warning signs

- Coughing, choking, colour change, or wet voice with feeds
- Poor growth, dehydration, prolonged meals, or marked fatigue
- Persistent vomiting, pain, or severe texture restriction
- Loss of feeding skill or recurrent respiratory illness
- Caregiver-child feeding interaction marked by distress

# Dressing and toileting readiness

## Dressing progression

- Helps by pushing an arm through a sleeve or lifting a foot
- Removes simple loose clothing
- Puts on some clothing with help
- Manages larger fasteners before smaller fasteners

## Toileting readiness

- Stays dry for increasing periods and recognizes body signals
- Can reach and sit safely on the toilet or potty
- Communicates the need and follows simple steps
- Family can offer calm, consistent practice without punishment

# Play as a developmental process

- Play integrates movement, sensation, communication, problem solving, imagination, emotion, and relationships
- Infants explore through sensory and social play
- Toddlers repeat actions, imitate routines, and begin symbolic play
- Preschool children create roles, stories, rules, and shared plans
- Play reveals strengths and participation more naturally than a highly directed test
- Nurses can demonstrate simple play activities that fit the family’s routines and resources

**The most useful play activity is safe, enjoyable, responsive, and easy to repeat.**

# Types of play across early childhood

| Play pattern | Typical features | What it supports |
|---|---|---|
| Social sensory play | Faces, voices, touch, songs, peek-a-boo | Attachment, attention, turn-taking, regulation |
| Exploratory play | Mouthing, banging, dropping, filling, emptying | Cause and effect, motor control, sensory learning |
| Functional play | Uses an object in its familiar way | Imitation, concepts, daily routines |
| Pretend play | Acts out roles or uses one object for another | Language, symbolic thought, emotion, flexibility |
| Parallel and associative play | Plays beside peers and begins sharing materials | Peer awareness, imitation, early cooperation |
| Cooperative rule-based play | Shared purpose, roles, turns, simple rules | Planning, self-control, negotiation, belonging |

# Play-based developmental observation

1

## Prepare

Choose safe familiar objects matched to age, culture, sensory needs, and motor access.

2

## Wait

Allow spontaneous exploration before demonstrating or asking questions.

3

## Join

Follow the child’s attention, imitate, take turns, and add one manageable challenge.

4

## Record

Describe the child’s actions, communication, support needed, and response to help.

# Integrated milestone profile: 2 months

| Domain | Examples most babies show | Nursing observation |
|---|---|---|
| Social-emotional | Calms when spoken to or picked up, looks at faces, smiles during interaction | Reciprocity, comfort, state regulation |
| Language | Makes sounds other than crying, reacts to loud sounds | Voice response and vocal variety |
| Cognitive | Watches a moving person, looks at a toy briefly | Visual attention and tracking |
| Motor | Holds head up on tummy, moves both arms and legs, opens hands briefly | Symmetry, tone, head control |

# Integrated milestone profile: 4 months

| Domain | Examples most babies show | Nursing observation |
|---|---|---|
| Social-emotional | Smiles to gain attention, chuckles, seeks continued interaction | Initiation and shared pleasure |
| Language | Coos, vocalizes back, turns toward caregiver voice | Sound exchange and hearing response |
| Cognitive | Opens mouth for feeding cue, watches own hands | Anticipation and visual attention |
| Motor | Steady head, forearm support, hand-to-mouth, holds and swings at toys | Head control, reach, symmetry |

# Integrated milestone profile: 6 months

| Domain | Examples most babies show | Nursing observation |
|---|---|---|
| Social-emotional | Knows familiar people, laughs, enjoys mirror image | Familiarity and shared enjoyment |
| Language | Takes vocal turns, blows raspberries, squeals | Vocal variety and reciprocity |
| Cognitive | Mouths objects, reaches for a wanted toy, signals fullness | Exploration and purposeful reach |
| Motor | Rolls tummy to back, pushes on straight arms, supports sitting with hands | Transitions, trunk control, symmetry |

# Integrated milestone profile: 9 months

| Domain | Examples most babies show | Nursing observation |
|---|---|---|
| Social-emotional | Responds to name, reacts to caregiver leaving, enjoys peek-a-boo | Social response and attachment behaviour |
| Language | Uses repeated syllable strings, raises arms to be picked up | Sound variety and gestures |
| Cognitive | Looks for a dropped object, bangs objects together | Object search and cause and effect |
| Motor | Gets into sitting, sits unsupported, transfers objects, rakes food | Transitions, balance, bilateral hand use |

# Integrated milestone profile: 12 months

| Domain | Examples most babies show | Nursing observation |
|---|---|---|
| Social-emotional | Plays an interactive game with caregiver | Turn-taking and shared enjoyment |
| Language | Waves, uses a parent name, understands a brief no | Gesture, word meaning, response |
| Cognitive | Puts objects in a container, searches for a hidden toy | Object permanence and purposeful action |
| Motor and adaptive | Pulls to stand, cruises, pincer grasp, drinks from held open cup | Mobility, grasp, feeding participation |

# Integrated milestone profile: 15 months

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Copies peers, shows an object, claps, shows affection | Imitation and shared interest |
| Language | Attempts one or two words, points for help, follows direction with gesture | Words, pointing, comprehension |
| Cognitive | Uses common objects correctly, stacks two objects | Functional use and construction |
| Motor and adaptive | Takes a few independent steps, finger-feeds | Balance, gait, hand-to-mouth control |

# Integrated milestone profile: 18 months

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Checks caregiver proximity, points to share interest, helps with dressing | Joint attention and autonomy |
| Language | Attempts at least three words, follows a one-step direction without gesture | Expression and comprehension |
| Cognitive | Copies simple chores, uses toys functionally | Imitation and purposeful play |
| Motor and adaptive | Walks independently, scribbles, tries spoon and open cup, climbs on furniture | Gait, hand use, self-care safety |

# Integrated milestone profile: 2 years

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Notices distress in others, checks caregiver reaction in a new situation | Empathy and social referencing |
| Language | Combines two words, points to pictures and body parts, uses varied gestures | Word combinations and comprehension |
| Cognitive | Uses two objects together, operates simple controls | Problem solving and functional play |
| Motor and adaptive | Runs, kicks, uses stairs with help, eats with a spoon | Coordination and independence |

# Integrated milestone profile: 30 months

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Plays beside and sometimes with peers, follows simple routines | Peer engagement and cooperation |
| Language | Uses about 50 words, combines words with an action, uses pronouns | Vocabulary and sentence development |
| Cognitive | Pretends, solves a simple problem, follows two-step directions, identifies a colour | Symbolic thought and planning |
| Motor and adaptive | Jumps with both feet, turns pages, twists lids, removes loose clothing | Coordination and self-care |

# Integrated milestone profile: 3 years

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Settles after separation and joins peer play | Recovery and participation |
| Language | Has back-and-forth conversation, asks questions, states first name | Reciprocity and intelligibility |
| Cognitive | Draws a demonstrated circle and understands a familiar danger warning | Imitation, concept learning, safety |
| Motor and adaptive | Strings large items, puts on simple clothing, uses a fork | Bilateral hand use and self-care |

# Integrated milestone profile: 4 years

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Uses pretend roles, seeks peers, comforts others, adjusts behaviour by setting | Imagination, empathy, flexibility |
| Language | Uses four-word sentences, describes a recent event, answers simple questions | Narrative and comprehension |
| Cognitive | Names colours, predicts a familiar story event, draws a person with parts | Memory, concepts, representation |
| Motor and adaptive | Catches a large ball, serves food with help, unbuttons, uses finger-thumb pencil grasp | Coordination and self-care |

# Integrated profile: age 5 years

| Domain | Examples most children show | Nursing observation |
|---|---|---|
| Social-emotional | Takes turns in games, performs for others, completes simple chores | Rules, confidence, responsibility |
| Language | Tells a two-event story, discusses a story, sustains conversation, recognizes rhyme | Narrative and phonologic awareness |
| Cognitive | Counts to 10, names some numbers and letters, uses time words, attends 5 to 10 minutes | School readiness and attention |
| Motor and adaptive | Hops on one foot, buttons some buttons, writes some letters in own name | Balance, dexterity, self-care |

MODULE 4

# Middle childhood

School-age development, learning, health habits, chronic conditions, and family partnership



# School-age development

## Expected progression

- Steady physical growth and improved coordination
- More logical thinking about concrete events
- Increasing independence in self-care and school tasks
- Peer acceptance and competence become important

## Variation to explore

- Learning, attention, language, hearing, or vision concerns
- Bullying, school avoidance, anxiety, or low mood
- Early or delayed pubertal signs
- Chronic illness effects on attendance, participation, and self-image

# School-age health promotion

- Assess nutrition, movement, sleep, oral health, immunization, and injury prevention
- Ask the child directly about school, friendships, strengths, worries, and safety
- Promote routines that support learning and emotional regulation
- Use growth trends and blood pressure correctly for age and size
- Coordinate with family, school health services, and the clinical team
- Support participation for children with chronic illness or disability

# Learning and behaviour concerns

## Assessment

- Define the concern across home and school settings
- Review developmental, medical, sleep, sensory, language, and psychosocial history
- Identify strengths and tasks the child can complete
- Screen for bullying, stress, trauma, and caregiver strain

## Response

- Avoid labelling behaviour before understanding function
- Arrange hearing, vision, and clinical evaluation as indicated
- Support reasonable adjustments and consistent routines
- Set a follow-up point and check whether the plan improved participation

# Nursing care for a school-age child

1

## Include

Speak with the child as well as the caregiver.

2

## Explain

Use concrete language, demonstration, and choices within safe limits.

3

## Rehearse

Let the child practise inhalers, insulin, wound care, or coping skills.

4

## Evaluate

Ask for teach-back and observe the skill in context.

# Case: school absence and abdominal pain

PATIENT SNAPSHOT

An 8-year-old has recurrent morning abdominal pain on school days. Examination and growth are reassuring. The child becomes quiet when classmates are mentioned, and the parent requests more laboratory tests.

## Clinical reasoning questions

- What physical and psychosocial data are still needed?
- How will you speak with the child privately and safely?
- Which findings would change the urgency of referral?
- How can the plan address symptoms, school participation, and family concern?

MODULE 5

# Adolescence

Puberty, identity, autonomy, confidential care, risk assessment, and mental health



# Adolescent growth and puberty

## Physical development

- Adolescence spans ages 10 to 19 in the WHO definition
- Pubertal timing and tempo vary widely
- Growth acceleration, body composition changes, and sexual maturation occur
- Tanner staging describes physical maturation and requires consent and privacy

## Clinical interpretation

- Assess the pattern over time and family history
- Explore nutrition, exercise, chronic illness, medication, and emotional health
- Avoid assumptions based on body appearance or chronological age
- Refer when development is unexpectedly early, delayed, rapid, or asymmetric

# Adolescent cognitive and psychosocial development

## Emerging capacities

- Abstract reasoning and future planning become more available
- Identity, values, relationships, and belonging take greater importance
- Autonomy grows while caregiver support remains protective
- Peer context can influence behaviour and health choices

## Nursing implications

- Assess understanding and decision-making for the specific choice
- Give accurate information without shame or exaggeration
- Connect immediate priorities to longer-term consequences
- Support protective relationships and realistic steps

# Confidential adolescent care

- Explain privacy and its limits before sensitive questions
- Spend part of the visit with the adolescent alone when appropriate and lawful
- Use neutral, inclusive language and ask rather than assume
- Clarify consent, capacity, documentation, and local policy
- Break confidentiality when there is a serious safety concern, using the least harmful pathway
- Tell the adolescent what will happen next whenever safety permits

**Local law and institutional policy determine specific consent and confidentiality rules.**

# HEEADSSS psychosocial interview

| Area | Examples of neutral opening questions |
|---|---|
| Home | Who lives with you? Where do you feel most comfortable or stressed? |
| Education / employment | How is school or work going? Any recent change in attendance or performance? |
| Eating | How do you feel about food, weight, and your body? |
| Activities | What do you enjoy? Who do you spend time with? |
| Drugs | What substances do people your age use? What has your experience been? |
| Sexuality | Are you in any romantic or sexual relationships? What support or protection do you need? |
| Suicide / depression | In the past weeks, have you felt hopeless or thought about hurting yourself? |
| Safety | Do you feel safe at home, school, work, online, and in relationships? |

# Adolescent mental health and suicide risk

## Warning signs

- Persistent mood change, withdrawal, agitation, hopelessness, or falling function
- Self-harm, suicidal thoughts, a plan, access to means, or recent attempt
- Substance use, trauma, violence, severe insomnia, or psychotic symptoms
- Sudden calm after acute distress may still require careful assessment

## Nursing response

- Ask directly and calmly about suicide and self-harm
- Do not leave a person at imminent risk alone
- Reduce access to lethal means and activate the emergency pathway
- Document risk, protective factors, consultation, handover, and follow-up

# Sexual and reproductive health

- Provide age-appropriate, evidence-based information and invite questions
- Assess relationships, consent, coercion, pregnancy possibility, infection risk, and access to care
- Use inclusive language for sexual orientation and gender identity
- Offer prevention, testing, vaccination, contraception counselling, and referral according to local policy
- Protect privacy and recognize safeguarding concerns
- Confirm understanding with teach-back

**Respect, accuracy, and confidentiality improve the chance that adolescents disclose what they need.**

# Case: declining school performance

PATIENT SNAPSHOT

A 16-year-old attends for headaches and fatigue. School performance has fallen. The parent answers every question. The adolescent avoids eye contact and asks whether the conversation is confidential.

## Clinical reasoning questions

- How will you explain confidentiality and arrange private time?
- Which HEEADSSS areas are most urgent?
- What questions assess depression and suicide risk directly?
- How will the plan include the adolescent while involving safe support?

MODULE 6

# Adulthood

Roles, relationships, reproductive health, prevention, caregiving, and adaptation to change

# Early adulthood

## Developmental themes

- Identity continues to develop through education, work, relationships, and community
- Intimacy and interdependence become central for many adults
- Health habits established here influence later risk
- Timelines differ; marriage, parenthood, and career are not universal markers

## Nursing priorities

- Mental health, sleep, substance use, sexual and reproductive health
- Occupational exposure, safety, and health literacy
- Preventive care based on individual risk and local guidance
- Respect for diverse family structures, roles, and goals

# Pregnancy, parenthood, and family transition

- Pregnancy and birth can change identity, relationships, work, sleep, and mental health
- Assess both practical support and emotional adjustment
- Include partners or family members only with the patient’s agreement
- Screen for violence, depression, anxiety, substance use, and unmet social needs
- Support responsive caregiving without presenting one family form as the norm
- Link families to community and clinical resources early

# Middle adulthood

## Common transitions

- Increasing responsibility at work or in the community
- Care for children, parents, partners, or relatives
- Menopause and other age-related physiologic changes
- Bereavement, changing relationships, or employment disruption

## Assessment priorities

- Cardiometabolic risk, cancer prevention, musculoskeletal health, and sleep
- Mood, substance use, caregiver burden, and social connection
- Function and symptoms rather than age stereotypes
- Readiness for change and access to sustainable support

# Adult preventive nursing

1

## Risk

Use age, history, exposures, family history, and current function.

2

## Guidance

Apply current national or institutional screening and immunization guidance.

3

## Choice

Explain benefits, limitations, possible harms, and alternatives.

4

## Follow-up

Track abnormal results and remove barriers to completion.

# Supporting life transitions

- Ask what the change means to the person before assuming distress or resilience
- Identify losses, new roles, practical barriers, and existing strengths
- Assess coping, sleep, mood, relationships, finances, and safety
- Use brief interventions and referrals that match readiness and resources
- Review progress because adaptation changes over time

**Development in adulthood includes gain, maintenance, adaptation, and loss.**

# Case: the exhausted family caregiver

PATIENT SNAPSHOT

A 47-year-old caring for a parent with stroke has uncontrolled blood pressure, poor sleep, and missed appointments. The patient says, “There is no time for me.”

## Clinical reasoning questions

- What developmental and social roles shape this presentation?
- Which immediate health risks require action?
- How will you assess caregiver strain and available support?
- What small plan is realistic this week, and how will it be reviewed?

MODULE 7

# Older adulthood

Healthy ageing, functional ability, frailty, cognition, medication safety, and person-centred goals

# Normal ageing and disease

## Expected age-related change may include

- Reduced physiologic reserve and slower recovery
- Changes in vision, hearing, sleep, muscle mass, and balance
- Slower processing speed with preserved ability to learn
- Different priorities shaped by health, roles, environment, and meaning

## Never explain away

- Acute confusion, new functional loss, falls, weight loss, pain, or incontinence
- Depression, isolation, abuse, neglect, or medication harm
- Sensory loss that limits communication or consent
- Symptoms that need assessment simply because the person is older

# Healthy ageing and functional ability

1

## Intrinsic capacity

Physical and mental capacities, including mobility, cognition, senses, vitality, and psychological health.

2

## Environment

Housing, transport, relationships, services, attitudes, and assistive resources.

3

## Interaction

Function emerges from the fit between capacity and environment.

4

## Nursing goal

Support what matters to the person and preserve meaningful activity.

# Delirium, dementia, and depression

| Feature | Delirium | Dementia | Depression |
|---|---|---|---|
| Onset | Acute, hours to days | Usually gradual | Weeks to months, sometimes after stress or loss |
| Course | Fluctuates during the day | Progressive or stepwise | Variable; may improve with treatment |
| Attention | Impaired | Often preserved early | May appear reduced because of low motivation |
| Consciousness | Often altered | Usually clear until late | Clear |
| Priority | Medical emergency: find the cause | Assess cognition, function, safety, and support | Assess suicide risk and treatable contributors |

# Comprehensive gerontological assessment

- Goals, values, decision-making support, and advance care preferences
- Basic and instrumental activities of daily living
- Mobility, falls, frailty, nutrition, continence, pain, sleep, and skin
- Cognition, mood, hearing, vision, and communication
- Medication review including adherence, interactions, and treatment burden
- Home safety, caregiver capacity, finances, transport, and social connection

# Nursing priorities in older adulthood

## Preserve ability

- Mobilize safely and avoid unnecessary bed rest
- Support nutrition, hydration, sleep, and sensory function
- Simplify routines and medication plans when possible
- Use aids and environmental changes to improve participation

## Prevent harm

- Detect delirium and acute decline early
- Reduce falls, pressure injury, medication harm, and deconditioning
- Recognize abuse, neglect, and caregiver exhaustion
- Plan transitions and follow-up with the person and caregivers

# Case: a sudden loss of independence

PATIENT SNAPSHOT

An 82-year-old who normally cooks and shops independently becomes confused after admission for a urinary infection. The family says, “This must be dementia.” The patient uses hearing aids that are at home.

## Clinical reasoning questions

- Which features suggest delirium?
- What reversible contributors will you assess first?
- How will you support orientation, hearing, mobility, sleep, and hydration?
- What baseline information and discharge risks must be documented?

MODULE 8

# Integrated lifespan nursing

Communication, culture, safeguarding, clinical reasoning, care planning, and performance assessment

# Communication across the lifespan

| Stage | Useful approach | Avoid |
|---|---|---|
| Infant | Observe cues; keep caregiver close; use soothing voice and pacing | Separating unnecessarily or ignoring caregiver knowledge |
| Young child | Use play, simple words, demonstration, and limited choices | Abstract explanations or false reassurance |
| School-age child | Explain sequence and sensations; involve the child in tasks | Speaking only to the adult |
| Adolescent | Clarify confidentiality; ask neutral questions; support autonomy | Assumptions, lectures, or avoidable breaches of privacy |
| Adult | Link care to goals, roles, readiness, and practical constraints | Treating age as a complete explanation |
| Older adult | Optimize hearing and vision; allow time; confirm preferences | Elderspeak or automatically directing decisions to family |

# Culture, family, and social context

## Ask with curiosity

- Who should be involved in decisions?
- What does healthy development mean in this family?
- Which routines, beliefs, or responsibilities affect care?
- What barriers make the plan difficult to follow?

## Practise cultural humility

- Do not treat culture as a checklist or predict beliefs from identity
- Use qualified interpreters when needed
- Respect family roles while protecting the patient’s voice and safety
- Adapt teaching and verify understanding

# Trauma-informed care and safeguarding

## Trauma-informed practice

- Promote physical and emotional safety
- Explain what will happen and ask permission when possible
- Offer choice, control, and predictable pacing
- Avoid repeated unnecessary retelling

## Safeguarding response

- Recognize concerning injuries, behaviour, neglect, exploitation, or coercion
- Respond calmly and avoid leading questions
- Document objective findings and exact words
- Follow local reporting, protection, and emergency procedures

# Clinical reasoning workflow

1

## Recognize

Compare current function with baseline, developmental expectations, and the patient’s context.

2

## Prioritize

Identify immediate threats, loss of skills, rapid change, or safeguarding risk.

3

## Respond

Assess, teach, adapt care, consult, refer, or escalate.

4

## Reassess

Confirm response, document the trajectory, and close the follow-up loop.

# Developmental nursing care plan

## Assessment statement

- Describe function and concern in observable terms
- Include strengths, trajectory, context, and patient or caregiver priorities
- Separate developmental variation from acute illness
- State the risk that nursing action can reduce

## Plan and evaluation

- Choose a measurable functional or safety outcome
- Adapt communication and intervention to developmental capacity
- Assign referral, teaching, documentation, and follow-up responsibilities
- Evaluate change in function, understanding, safety, or participation

# Suggested OSCE stations

- Plot and interpret serial growth measurements for an infant
- Explain a developmental referral to the parent of an 18-month-old
- Conduct the opening of a confidential adolescent interview
- Teach inhaler use to a school-age child and caregiver
- Assess an older adult with acute confusion and hearing impairment
- Handover a safeguarding concern using objective, precise language

**Each station should assess communication, safety, clinical reasoning, and documentation.**

# Clinical performance rubric

| Domain | Beginning | Competent | Excellent |
|---|---|---|---|
| Assessment | Collects incomplete or poorly prioritized data | Collects relevant developmental and clinical data | Integrates trajectory, function, context, and strengths |
| Communication | Uses generic or age-inappropriate language | Adapts language and checks understanding | Promotes autonomy, trust, and shared decisions |
| Safety | Misses warning signs or escalation | Recognizes risk and follows the pathway | Anticipates harm and closes the follow-up loop |
| Care planning | Lists tasks without measurable outcomes | Links actions to a clear outcome | Builds a feasible plan with patient, family, and team |
| Documentation | Vague or judgmental | Objective, complete, and timely | Captures baseline, change, decisions, and responsibilities |

# High-yield review

## Always remember

- Trend is more informative than one measurement
- Loss of skills or acute functional decline needs prompt action
- Screening identifies risk; it does not establish a diagnosis
- Communication must match capacity, sensory needs, and context

## Ask in every stage

- What can this person do now, and what has changed?
- What matters to the person and family?
- What threatens safety or participation?
- Who owns the next step, and when will it be reviewed?

# Course takeaways

- Human development continues from conception through the end of life
- Age provides context, but function and trajectory guide nursing judgement
- Biology, relationships, environment, and time interact
- Accurate measurement and observation make early action possible
- Developmentally appropriate communication improves safety and trust
- Nurses connect assessment to teaching, protection, referral, and follow-up

**The goal is not to memorize a “normal person.” The goal is to recognize patterns and respond safely.**

# Core references


- World Health Organization. Child Growth Standards and growth assessment training resources
- World Health Organization. Adolescent health and healthy ageing resources
- Centers for Disease Control and Prevention. Learn the Signs. Act Early. milestone resources
- American Academy of Pediatrics. Bright Futures Guidelines and Periodicity Schedule
- Hockenberry, Wilson, and Rodgers. Wong’s Nursing Care of Infants and Children
- Potter, Perry, Stockert, and Hall. Fundamentals of Nursing
- Touhy and Jett. Ebersole and Hess’ Gerontological Nursing and Healthy Aging